

Entregable Técnico: Nmap

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1. Objetivo del entregable y materiales necesarios

En este entregable voy a experimentar con la herramienta Nmap. Esta herramienta se emplea principalmente para el descubrimiento de equipos en una red, el escaneo de puertos y la enumeración de servicios, permitiendo obtener información sobre los dispositivos conectados, los puertos que tienen abiertos y los servicios que se encuentran en ejecución. A lo largo de esta práctica se utilizarán distintos comandos de Nmap para comprender su funcionamiento y analizar la información que proporciona.

En Ubuntu se necesita primero instalar esta herramienta, esto se puede realizar con `apt install nmap` (actuando como root con `sudo`), no sin antes hacer un update para asegurar instalar la versión más reciente de nmap:

```
sergio@sergio:~$ sudo apt update && sudo apt install nmap
[sudo] contraseña para sergio:
Lo siento, pruebe otra vez.
[sudo] contraseña para sergio:
Obj:1 http://es.archive.ubuntu.com/ubuntu noble InRelease
Des:2 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Des:3 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Des:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Des:5 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1.096 kB]
Des:6 http://es.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [270 kB]
Des:7 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Des:8 http://es.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17,7 kB]
Des:9 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1.229 kB]
Des:10 http://es.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [278 kB]
Des:11 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1.660 kB]
Des:12 http://es.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [327 kB]
Des:13 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Des:14 http://es.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34,9 kB]
Des:15 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [11,8 kB]
Des:16 http://es.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Des:17 http://es.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5.776 B]
Des:18 http://es.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10,5 kB]
Des:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [835 kB]
Des:20 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [188 kB]
Des:21 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [44,9 kB]
Des:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11,9 kB]
Des:23 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1.139 kB]
Des:24 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [260 kB]
Des:25 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1.176 kB]
Des:26 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [231 kB]
Des:27 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76,2 kB]
Des:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [24,2 kB]
Des:29 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [10,1 kB]
Descargados 9.885 kB en 3s (3.781 kB/s)
```

Localhost representa al propio equipo, equivale siempre a la IP 127.0.0.1, esto en Linux está configurado en el archivo `/etc/hosts`, que es un archivo que realiza prioritariamente la resolución local de cualquier dominio establecido dentro de este archivo a la IP que haya a su lado, siempre esta localhost con 127.0.0.1 al instalar Ubuntu.

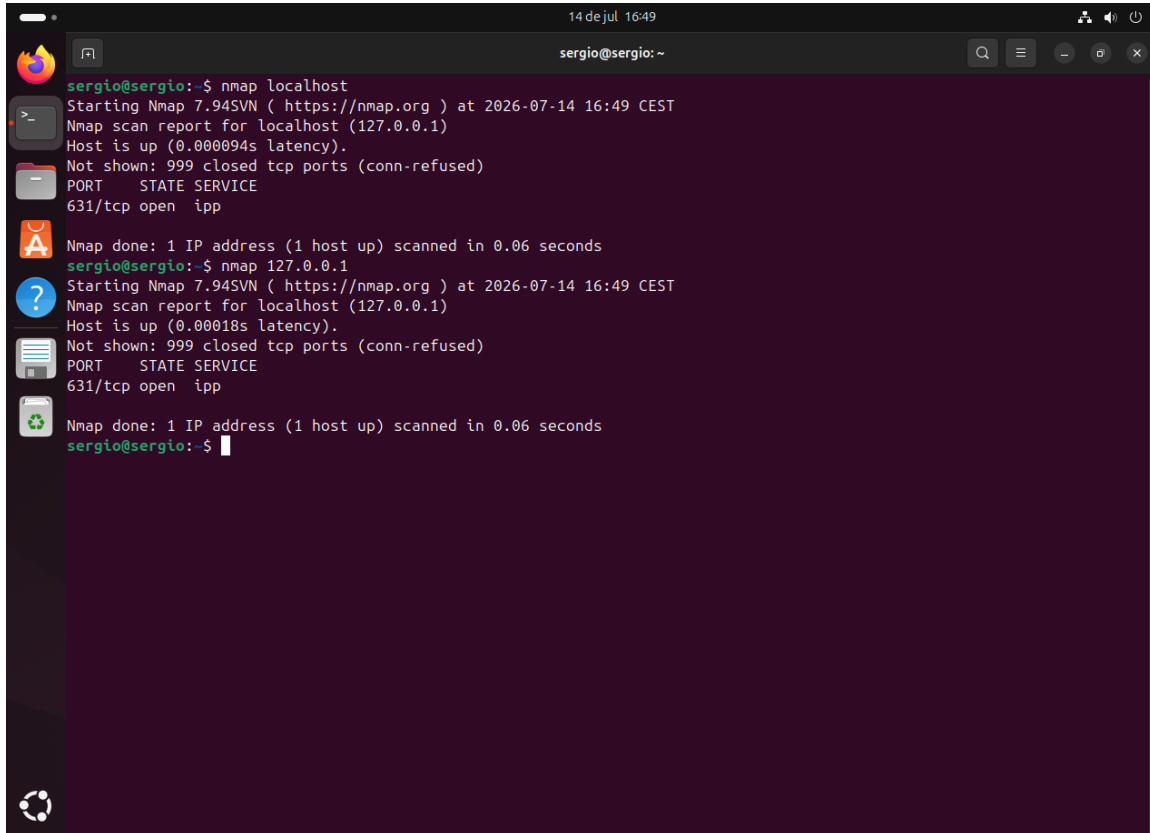


```
127.0.0.1 localhost
127.0.1.1 sergio

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

La IP 127.0.0.1 no está asignada a ninguna interfaz física del equipo, sino que es una dirección especial perteneciente al rango de loopback (127.0.0.0/8). El tráfico dirigido a esta dirección nunca abandona el sistema operativo, solo se procesa internamente.

Podemos ver que al escanear con Nmap localhost y 127.0.0.1 sale el mismo resultado, se tiene abierto el puerto TCP 631 en el equipo y el resto escaneados están cerrados, aunque no se muestra mucha más información. Además, solo se han escaneado los 1000 puertos más comunes asociados a servicios conocidos (21,22,23,25,53...).

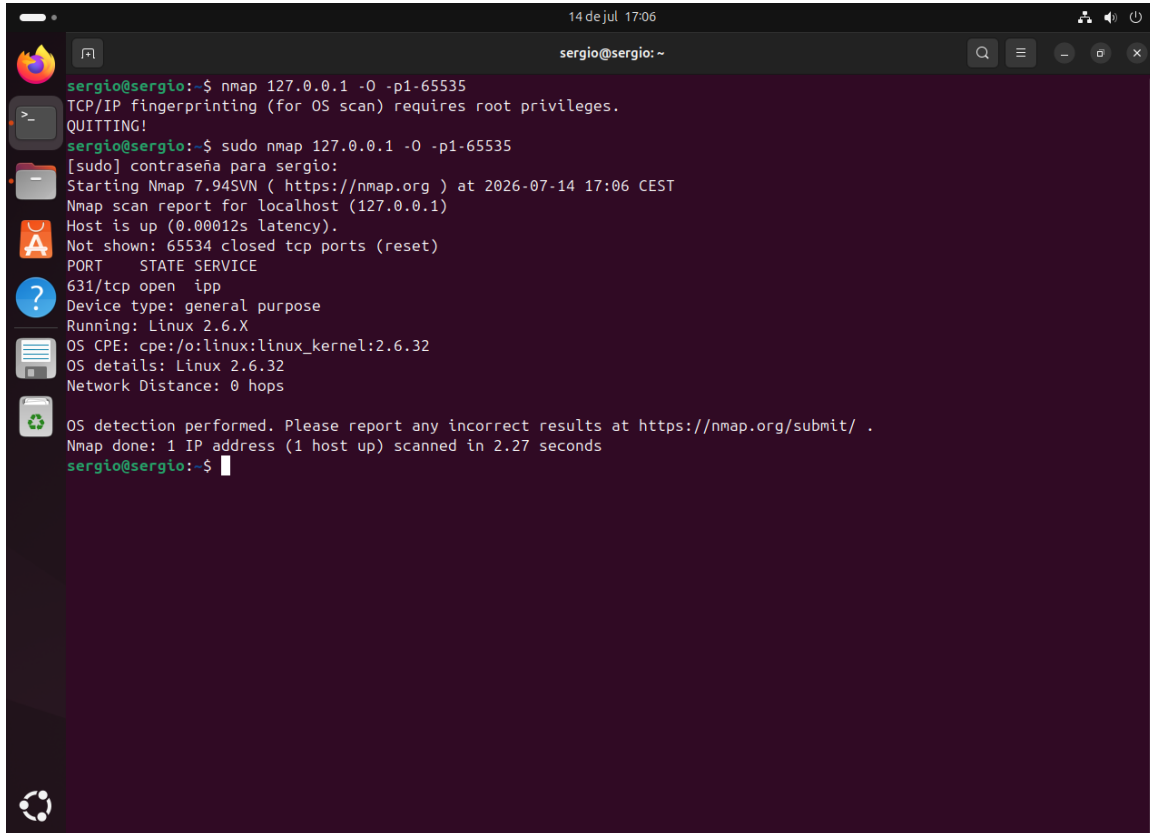


```
sergio@sergio:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 16:49 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000094s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
631/tcp   open ipp

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
sergio@sergio:~$ nmap 127.0.0.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 16:49 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00018s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
631/tcp   open ipp

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
sergio@sergio:~$
```

Nmap tiene varias opciones para aumentar el rango del escaneo, por ejemplo con la opción `-O` escaneará el SO del equipo y con `-p1-65535` o `-p-` escaneará todos estos puertos, que son todos los de TCP (Se pueden establecer distintos rangos).

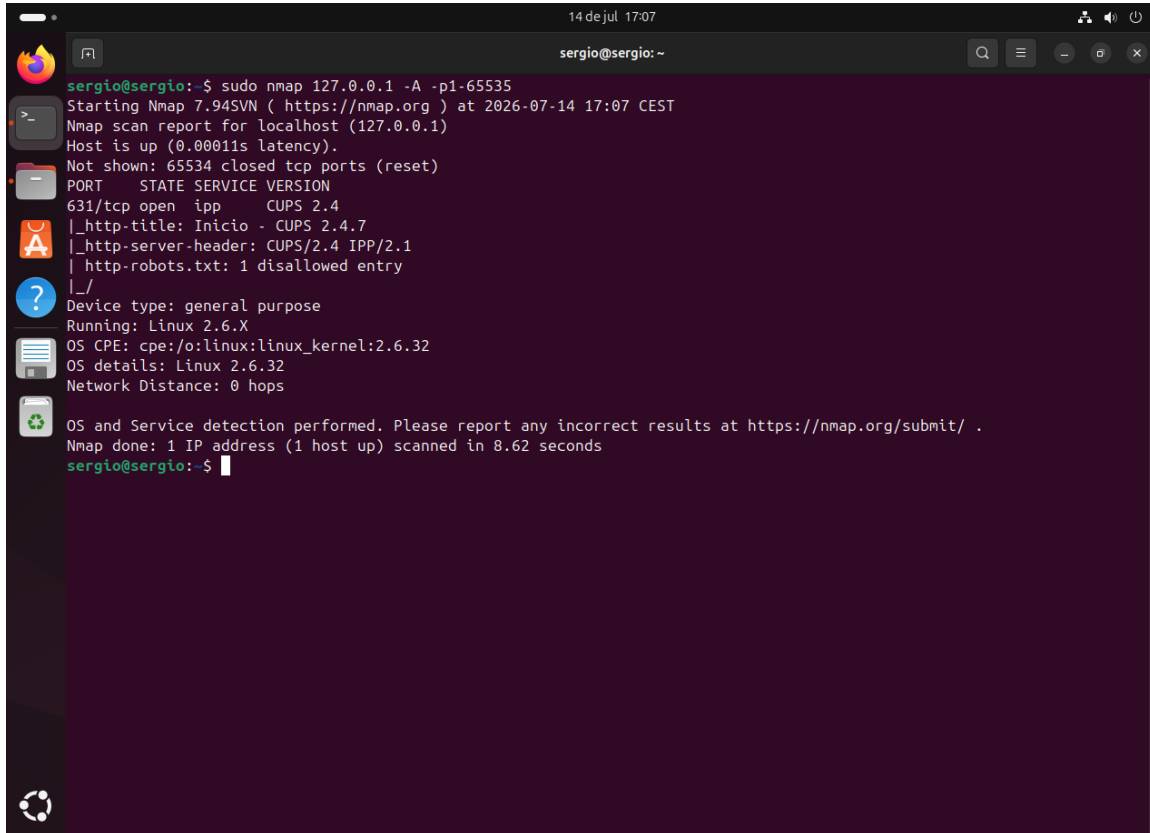


A terminal window titled "sergio@sergio: ~" with a search bar and window controls. The terminal shows the execution of nmap on localhost (127.0.0.1) with the -O flag. The first attempt fails due to lack of root privileges. The second attempt, using sudo, succeeds and displays a detailed scan report. The report includes the nmap version (7.94SVN), scan time (2026-07-14 17:06 CEST), host status (up), and a list of open ports (631/tcp). It also provides OS detection details (Linux 2.6.32) and a network distance of 0 hops. The terminal ends with the prompt "sergio@sergio:~\$".

```
sergio@sergio:~$ nmap 127.0.0.1 -O -p1-65535
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!
sergio@sergio:~$ sudo nmap 127.0.0.1 -O -p1-65535
[sudo] contraseña para sergio:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 17:06 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE
631/tcp   open ipp
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.27 seconds
sergio@sergio:~$
```

También con la opción `-A` se muestra se realiza un escaneo mucho más completo, por ejemplo me dice a que servicio esta asociado el puerto abierto, el 631 pertenece a CUPS.

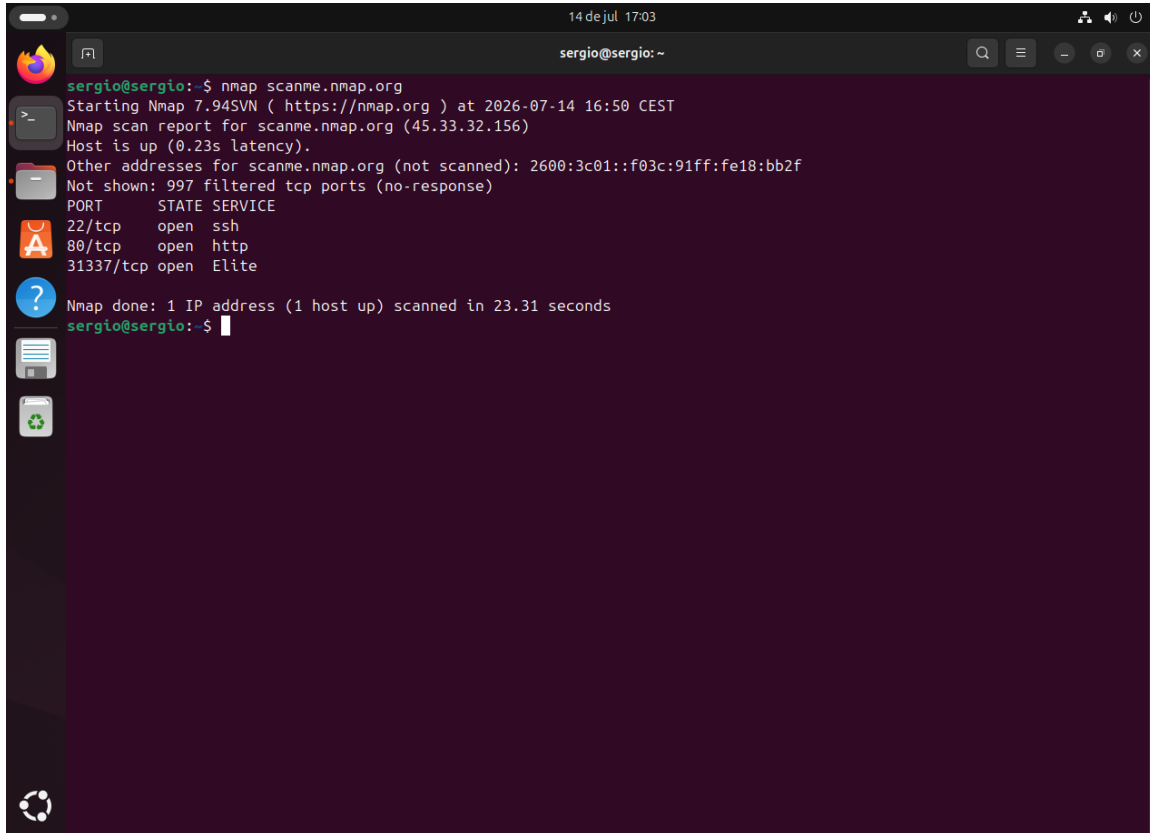


```
sergio@sergio:~$ sudo nmap 127.0.0.1 -A -p1-65535
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 17:07 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
631/tcp   open  ipp      CUPS 2.4
|_http-title: Inicio - CUPS 2.4.7
|_http-server-header: CUPS/2.4 IPP/2.1
|_http-robots.txt: 1 disallowed entry
|_/
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32
OS details: Linux 2.6.32
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.62 seconds
sergio@sergio:~$
```

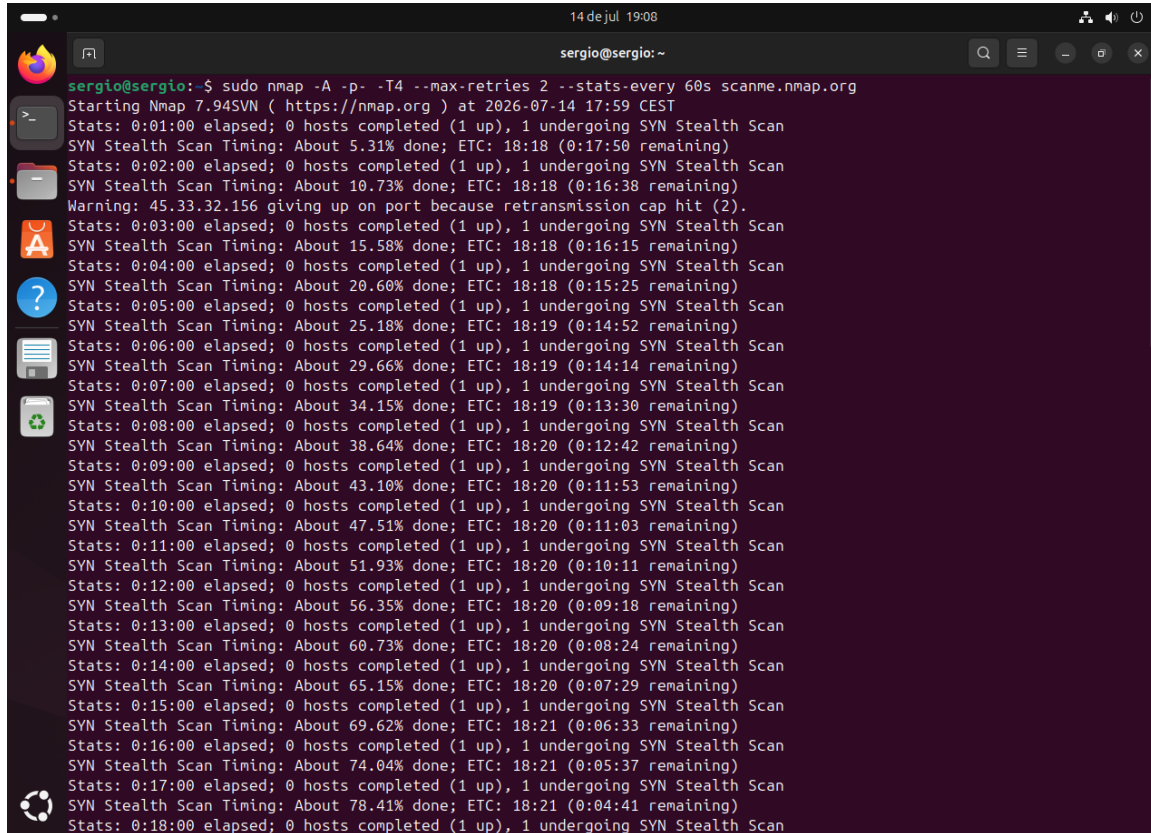
3. Nmap a scanme.nmap.org

scanme.nmap.org es un servidor público mantenido por los desarrolladores de Nmap para que cualquiera pueda practicar escaneos de forma legal. Este es el resultado de un nmap simple a este dominio, mostrando el puerto ssh, http y Elite abiertos y el resto escaneados cerrados.

A terminal window titled 'sergio@sergio: ~' with a dark purple background. The window shows the output of an Nmap scan command. The text is as follows:

```
sergio@sergio:~$ nmap scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 16:50 CEST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
31337/tcp  open  Elite
Nmap done: 1 IP address (1 host up) scanned in 23.31 seconds
sergio@sergio:~$
```

También realizo un Nmap completo con la opción -A y el escaneo de todos los puertos a este dominio, aunque tarda bastante tiempo en completarse, por lo que también añadí la opción de -T4 que reduce los tiempos de espera además de --max-retries 2 que solo realiza dos reintentos en cada escaneo de puerto en vez de los 10 por defecto, por último agregué --stats-every 60s para que cada minuto avise del porcentaje completado hasta ese momento por nmap.

A terminal window titled 'sergio@sergio: ~' with a search bar and window controls. It shows the execution of 'sudo nmap -A -p- -T4 --max-retries 2 --stats-every 60s scanme.nmap.org'. The output displays the progress of a SYN Stealth Scan, with statistics every minute. It shows 0 hosts completed and 1 host undergoing the scan. A warning is issued at 0:03:00 for IP 45.33.32.156. The scan progresses to 78.41% done at 0:17:00.

```
sergio@sergio:~$ sudo nmap -A -p- -T4 --max-retries 2 --stats-every 60s scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-14 17:59 CEST
Stats: 0:01:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.31% done; ETC: 18:18 (0:17:50 remaining)
Stats: 0:02:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 10.73% done; ETC: 18:18 (0:16:38 remaining)
Warning: 45.33.32.156 giving up on port because retransmission cap hit (2).
Stats: 0:03:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 15.58% done; ETC: 18:18 (0:16:15 remaining)
Stats: 0:04:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.60% done; ETC: 18:18 (0:15:25 remaining)
Stats: 0:05:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 25.18% done; ETC: 18:19 (0:14:52 remaining)
Stats: 0:06:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 29.66% done; ETC: 18:19 (0:14:14 remaining)
Stats: 0:07:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 34.15% done; ETC: 18:19 (0:13:30 remaining)
Stats: 0:08:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.64% done; ETC: 18:20 (0:12:42 remaining)
Stats: 0:09:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 43.10% done; ETC: 18:20 (0:11:53 remaining)
Stats: 0:10:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 47.51% done; ETC: 18:20 (0:11:03 remaining)
Stats: 0:11:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 51.93% done; ETC: 18:20 (0:10:11 remaining)
Stats: 0:12:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 56.35% done; ETC: 18:20 (0:09:18 remaining)
Stats: 0:13:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 60.73% done; ETC: 18:20 (0:08:24 remaining)
Stats: 0:14:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 65.15% done; ETC: 18:20 (0:07:29 remaining)
Stats: 0:15:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 69.62% done; ETC: 18:21 (0:06:33 remaining)
Stats: 0:16:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 74.04% done; ETC: 18:21 (0:05:37 remaining)
Stats: 0:17:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 78.41% done; ETC: 18:21 (0:04:41 remaining)
Stats: 0:18:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```

Tardó aproximadamente 35 minutos y dió el siguiente resultado, mostrando servicio, versiones, traceroute etc.

```

sergio@sergio: ~
14 de jul 19:08

Stats: 0:33:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 18:32 (0:00:00 remaining)
Stats: 0:34:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 18:33 (0:00:00 remaining)
Stats: 0:35:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 18:34 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.044s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 56163 closed tcp ports (reset), 9368 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|_ 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_ 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Aggressive OS guesses: DD-WRT v24-sp2 (Linux 2.4.37) (99%), Actiontec MI424WR-GEN3I WAP (99%), Linux 3.2 (96%), Linux 4
.4 (96%), Microsoft Windows XP SP3 or Windows 7 or Windows Server 2012 (93%), Microsoft Windows XP SP3 (91%), BlueArc T
itan 2100 NAS device (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1  0.30 ms _gateway (192.168.63.2)
2  0.31 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2124.48 seconds
sergio@sergio: $

```

4. Conclusión y aprendizaje

En esta práctica he empleado Nmap para hacer diversas pruebas, esta es una herramienta útil para escaneo de puertos en la red principalmente. En general he aprendido diversas opciones que ofrece este comando en Linux.